



New Maisine[®] CC

A unique pharmaceutical oil for solubility and oral bioavailability enhancement



People make our name



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This white paper features Maisine® CC, Gattefossé's tradename for glyceryl monolinoleate (USP-NF), also known as corn oil mono-, di- and triglycerides. This unique pharmaceutical oil has a long and successful history of use in lipid-based formulation (LBF) for oral bioavailability enhancement.

In LBF Maisine® is associated with drug solubilization and increased oral bioavailability and in certain cases mitigation of food effect and lymphatic drug absorption - thereby reducing first-pass metabolism.

LBF is a viable approach for BCS Class II drugs, particularly those with high Log P and good solubility in lipid excipients. LBF are classified in 4 types (Pouton, 2000 and 2006), depending on the functionality of excipients – oily vehicle, water soluble or insoluble surfactants and/or co-solvents – and their respective concentration. Each type has advantages and drawbacks as summarized in [CALLOUT BOX 1](#).

Maisine® is the oily vehicle of choice in Type I, II and III LBF due to its physico-chemical properties and regulatory status.

CALLOUT BOX 1: Lipid-based Formulation at a glance (Pouton, 2006)

	Type I	Type II	Type IIIA	Type IIIB	Type IV
Composition	Oily vehicle: Maisine® CC	100%	40 - 80	40 - 80	<20
	Water insoluble surfactants		20 - 60		0 - 20
	Water soluble surfactants			20 - 40	20 - 50
	Hydrophilic co-solvents			0 - 40	20 - 50
Particle size of the dispersion (nm)		Coarse	100 - 250	100 - 250	50 - 100
Characteristics <i>in vitro</i>		Non-dispersing; requires digestion	SEDDS Turbid o/w dispersion	SEDDS/SMEDDS Almost clear dispersion	SMEDDS Clear dispersion
Possible advantages		GRAS status; simple; mitigate food-effect; induce lymphatic transport	Unlikely to lose solvent capacity on dispersion; digestion can help; mitigate food-effect; induce lymphatic transport	Digestion can help; mitigate food-effect; induce lymphatic transport	Drug absorption without digestion
Possible disadvantages		Only for highly lipophilic drug	For highly lipophilic drugs	Possible loss of solvent capacity on dispersion <i>in vivo</i>	Likely loss of solvent capacity <i>in vivo</i> ; less easily digested



ASK FOR

Our White paper on lipid-based formulations

Researching indications in lipid-based formulations: from effective prediction to successful development

CONCLUSIONS

Pharmaceutical industry's biggest challenge is the effective delivery of poorly soluble drugs. Many molecules are studied for potential formulation using lipid-based formulations (LBF). From the early days of these formulations in the 1950s, LBF have been extensively studied and optimized, allowing greater formulation flexibility and increasing the number of drugs that can be formulated. However, only a few products have reached the market due to a lack of predictive factors to describe the formulation. Some collaboration with lipid-based formulations in the last decade has led to the development of novel products for oral bioavailability enhancement. Together with Gattefossé's knowledge and expertise in lipid-based formulations, industry can now rely on solid evidence for successful drug development.

INTRODUCTION

Development of orally active drugs for oral administration has many associated challenges for formulators including the need to be able to achieve drug solubility in the physiological environment of gastrointestinal tract (GIT). This is one of the major hurdles for BCS Class II drugs that must be overcome to ensure adequate absorption and systemic availability.

It is well known that the impact of new Chemical Entities (CEs) in the drug development pipeline and novel active compounds for the pharmaceutical industry's interest in finding novel drugs and solutions to improve the oral bioavailability of new drugs is increasing.

However, as a number of functional lipid excipients and lipid-based formulations is available, it is a challenge to determine the best formulation for a given drug. The use of lipid-based formulations (LBF) has been studied - often in collaboration with academia - as a fundamental research program, involving the characterization of individual critical material attributes, in order to develop and optimize formulation applications.

The white paper provides an overview of the lipid-based formulation approach that has been applied and leading to the development of lipid-based formulations in the pharmaceutical industry and to new products. This white paper provides more successful drug development.

Maisine® CC

- New grade: optimized grade of Maisine® 35 – 1
- Winterized: crystal clear liquid at room temperature
- Same composition in mono-, di- and triglycerides
- Same solubilizing performance with easier handling

Maisine® CC is manufactured according to GMP by Gattefossé, certified by ANSM (the French Health Authority). The winterization process delivers highly reproducible Maisine® CC composition with guaranteed content in mono-, di- and triglycerides and very low impurity levels.

Crystal Clear Advantages of Maisine® CC

When developing a lipid-based formulation Maisine® CC is the oily vehicle of choice:

- **Regulatory approval:** Maisine® CC meets the current specifications of the European monograph for glycerol monolinoleate and the US pharmacopoeia for glyceryl monolinoleate.
- **A long history of use:** Corn glycerides have previously been used in authorized oral medicinal products (capsule, soft gelatin capsule, oral solution). The most well-known formulation containing corn glycerides is Neoral®, with cyclosporine A.
- **Excellent solubilizing capacity:** Its composition in mono- and diglycerides enables greater solubilization than standard triglyceride oils.
- **Long chain digestible glyceride** promoting lymphatic absorption
- **A good miscibility** with other excipients
- **Compatibility** with hard and soft capsule shell
- **A limpid liquid** at room temperature, with facilitated handling and solubility screening
- **A high purity** excipient with good chemical stability

The complete physicochemical and functional properties of Maisine® CC are listed in [CALLOUT BOX 2](#).

CALLOUT BOX 2: Maisine® CC identity card



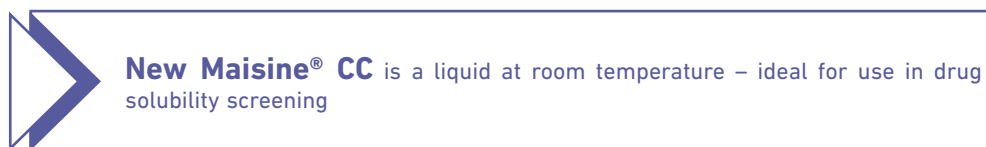
Names		
Gattefossé tradename	Maisine® CC	
EP	glycerol monolinoleate	
USP-NF	glyceryl monolinoleate	
INCI	glyceryl linoleate	
Other denominations	corn glycerides mixture of mono-, di- and triglycerides derived from corn oil	
CAS	68424-61-3 (or 26545-74-4) (or 91744-20-6)	
EINECS	270-312-1 (or 247-788-4) (or 294-582-5)	
Description		
Physical form	Limpid, yellow liquid, color <8.0 (Gardner scale)	
Composition	Total monoglycerides content	32.0 to 52.0 %
	Total diglycerides content	40.0 to 55.0 %
	Total triglycerides content	5.0 to 20.0 %
Regulatory		
Product conforms to EP, USP/NF pharmacopeias US DMF Type IV N° 5797		
Precedence of use (FDA Inactive Ingredient Database)		
Corn glycerides have previously been used in authorized oral medicinal products (capsule, soft gelatin capsule, oral solution and tablet). FDA IID: corn glycerides, UNI: 1DAF35W3S2		
Functions		
Solubilizer for lipophilic drugs	Mitigates food effects	
Enhancer for oral bioavailability	Promotes lymphatic absorption	
Physicochemical properties		
HLB	1	
Viscosity at 20°C	120 mPa.s	
Surface tension	32.2 ± 0.1 mN/m	
Compatible with capsule shell		
Solubility		
Ethanol 96°	Very soluble	
Chloroform	Soluble	
Methylene chloride	Soluble	
n-Hexane	Insoluble	
Water	Insoluble	
Vegetable oils	Very soluble	
Mineral oils	Very soluble	



Maisine® CC: high performance oil in LBF type I, II and III

LBFs are designed to solubilize poorly soluble drugs and increase oral bioavailability. The excipients constituting the LBF are selected for their capacity to solubilize the entire therapeutic drug dose in the dosage unit and to maintain drug solubility in the gastro-intestinal tract until absorption.

Maisine® CC is associated with improving bioavailability via several mechanisms as schematized in Figure 1.



Enhancing oral bioavailability

Bioavailability enhancement is achieved when the excipients used in the LBF are able to provide effective solubilization of the drug in the dosage form and maintain the drug in a solubilized state *in vivo* during digestion and absorption.

Most LBFs are multi-component systems which are carefully designed to provide and maintain drug solubility. There are numerous studies describing the use of Maisine® in LBFs which are associated with improved oral bioavailability (Table 1).

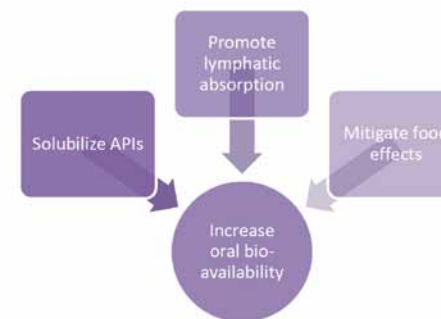


Figure 1: Benefits of Maisine® CC in lipid-based formulations

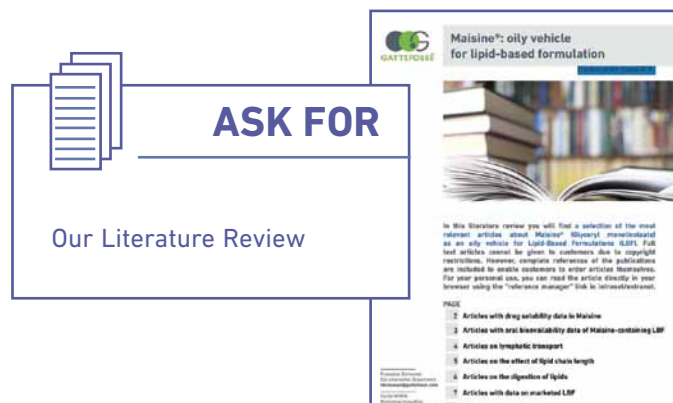
Table 1: Oral bioavailability enhancement in LBF containing Maisine®

Drug	Optimized LBF composition	Species	Reference
Atazanavir	Maisine® 46% Transcutol® 54%	Rat	Singh and Pai, 2014
Ezetimibe	Maisine® 36% Labrasol® 64%	Rat	Bandhyopadhyay <i>et al.</i> , 2012
Indirubin	Maisine® 15% Cremophor® EL 40% Transcutol® 45%	Rat	Chen <i>et al.</i> , 2012
Lopinavir	Maisine® 30% Tween 80 43% Transcutol® 27%	Rat	Garg <i>et al.</i> , 2016
Nelfinavir mesylate	Maisine® 20% Tween 80 50% Transcutol® 30%	Rabbit	Kamboj and Rana, 2016
Simvastatin	Maisine® 12% Plurol® oleique 11% Labrasol® 75% Transcutol® 1%	Rat	Singla <i>et al.</i> , 2009
Vinpocetine	Maisine® 10% Cremophor® EL 50% Transcutol® 40%	Rabbit	Abuhrama <i>et al.</i> , 2010

Enhancing drug solubility

The specific composition of Maisine® CC — mono and di glycerides being predominant — means it is a better solubilizer than standard triglyceride oils. Table 2 shows experimental data of solubility in Maisine® or in soybean oil for different lipophilic APIs.

There is an abundance data of drug solubility in Maisine in the scientific literature. The excellent solvent capacity of this oil makes it an essential excipient to screen in LBF formulation development.



A recent in-house study with cinnarizine (poorly soluble, high LogP = 5.8, therapeutic dose 25 mg, BCS class 2) showed that the addition of Maisine® to an LBF increased the solubility of drug during the digestion phase of the *in vitro* lipolysis test.

Type III LBF without Maisine®	Type III LBF with Maisine®
90% Labrasol® ALF	75% Labrasol® ALF
10% Capryol™ 90	10% Capryol™ 90
	15% Maisine® CC

Table 2: Data of experimentally determined equilibrium solubility of some API in Maisine® and soybean oil at 37°C (from Alskär et al., 2015)

API	Log P	Solubility* in Maisine® (mg/g)	Solubility* in soybean oil (mg/g)
Halofantrine	8.2	71.2	52.8
Clofazimine	6.9	11.7	9.5
Clotrimazole	5.2	75.4	15.1
Danazol	4.9	14.7	3.9
Ethinylestradiol	4.9	22.1	13.4
Disulfiram	4.6	49.0	21.4
Indomethacin	4.2	13.0	2.0
Perphenazine	4.2	92.5	16.7
Tolfenamic acid	4.1	14.1	3.8
Saquinavir	3.9	>123	2.3
Felodipine	3.6	37.7	9.6
Progesterone	3.6	66.2	30.4
Diflunisal	3.1	31.6	7.9
Naproxen	2.8	19.5	5.5
Carbamazepine	2.7	30.0	1.4

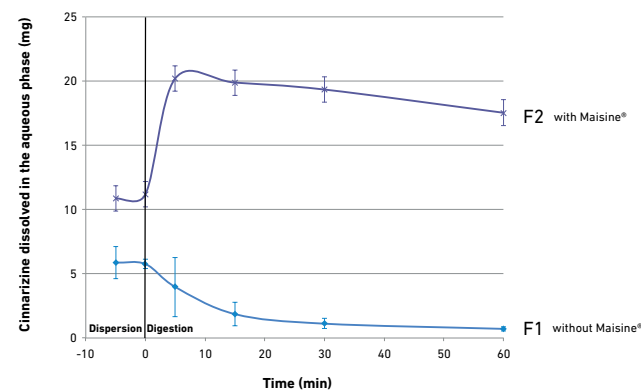


Figure 2: Maisine® increases drug solubility on dispersion and during digestion

Mitigating food effect

Some drug formulations are associated with variable bioavailability when administered with food (i.e. food effect). This is due to the physiological digestion of lipids in food and their ability to alter drug solubility. This can be a problem in drug development. Due to its composition, Maisine® can mitigate food effect.

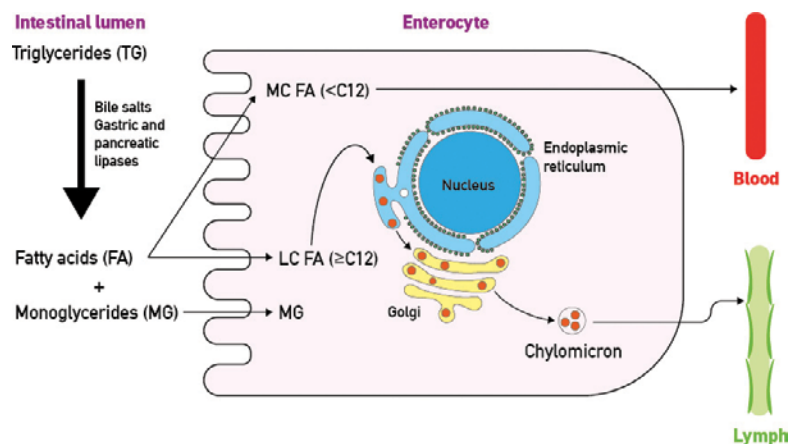
In 1993, Charman et al. published data on the food effect observed on danazol oral bioavailability in humans using a lipid (glycerol monooleate) emulsion and a non-lipidic formulation (market capsule). It showed that LC monoglycerides mitigate the food effect (Table 3). The authors concluded that LC-monoglycerides enable better intrinsic emulsification without the necessity of lipid hydrolysis compared with LC-triglycerides.

Table 3: LC monoglycerides mitigate food effect

	Market product Fasted state	Market product Fed state	Lipid emulsion Fasted state	Lipid emulsion Fed state
C_{max} (ng/mL)	37 ± 16	101 ± 42	155 ± 55	126 ± 56
T_{max} (h)	2.5 ± 1.4	3.1 ± 1.2	3.5 ± 1.2	4.9 ± 1.4
AUC^{0-36} (ng.h/mL)	204 ± 125	639 ± 259	779 ± 189	844 ± 194

Promoting lymphatic drug absorption

Maisine® CC is a long chain lipid ($C_{18:2}$) and as such is absorbed via the lymphatic route.



Two distinct studies compared the lymphatic transport of halofantrine with long chain triglycerides (LCT), compared to medium (MCT) and short chain (SCT) triglycerides, *in vivo* in rats (Caliph, 2000) and in dogs (Khoo, 2003). The results of these studies shown in Figure 3 and Figure 4 respectively clearly show that LCT are preferentially absorbed via the lymphatic system.

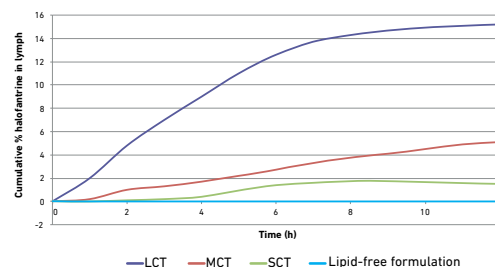


Figure 3: Halofantrine recovery in lymph in rats (from Caliph, 2000)

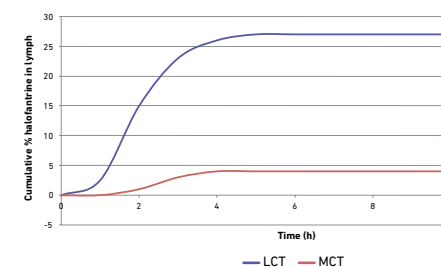


Figure 4: Halofantrine recovery in lymph in dogs (From Khoo, 2003)

Maisine®, a must-have excipient

Corn oil mono-, diglycerides or Glyceryl monolinoleate (Maisine®) has been used in marked approved drugs and is a widely studied excipient in LBF.

New Maisine® CC has all the benefits of a unique and well-characterized composition that delivers solubilizing properties superior to other oily vehicles – as such it is the pharmaceutical oil of choice in lipid-based formulation. Numerous studies demonstrate Maisine®'s advantages in increasing bioavailability, mitigating food effect to delivery consistent bioavailability and promoting lymphatic uptake.

Maisine® CC is a 'crystal clear' liquid at room temperature, ideal for handling and solubility screening.

Maisine® CC is available from Gattefossé along with a full range of functional excipients for LBF (Figure 5) together with technical support for LBF development. For more information please contact your local Gattefossé office: www.gattefosse.com.

Maisine® CC samples, product brochure and Formulation Guidelines for Solubility and Bioavailability Enhancement are available via your Gattefossé representative.

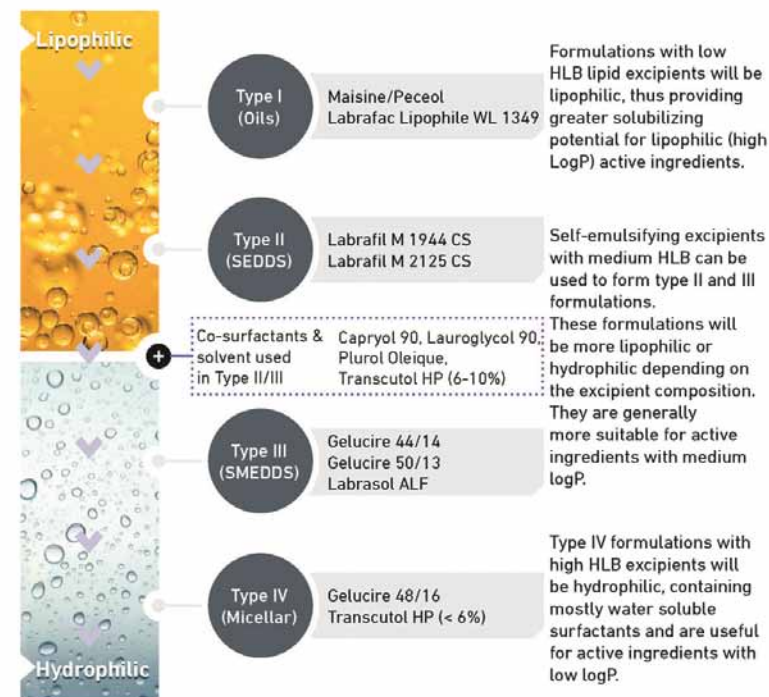


Figure 5: Gattefossé's full range of functional excipients for lipid-based formulations

www.gattefosse.com

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